

Paper 03 — Third-Party Plugin Control

FabFilter, iZotope, Waves and the Complete Plugin Ecosystem

WHYCREMISI RESEARCH PAPERS — N.03

Universal Third-Party Plugin Control

"Every knob of every plugin — accessible by AI."

1. The Digital Silo Problem

Audio plugins are isolated islands. FabFilter Pro-Q3 does not know what iZotope Ozone is doing. Waves SSL does not know the track level. WhyCremisi is the first system that **breaks these silos** — creating a unified control plane on which AI can operate the entire ecosystem.

2. Control Mechanisms

2.1 VST3 Parameter Automation (Primary)

VST3 exposes every plugin parameter as a normalised float [0.0–1.0] accessible from the host.

```
// ParameterMapper — conceptual example
void ParameterMapper::setPluginParameter(
    int trackId, const std::string& pluginName,
    const std::string& paramName, float value)
{
    auto* proc = getProcessorOnTrack(trackId, pluginName);
    int idx = lookupParamIndex(pluginName, paramName);
    proc->setParameterNotifyingHost(idx, value);
}
```

2.2 MIDI CC — Universal fallback for any MIDI-Learn capable plugin

2.3 OSC — Direct communication with Reaper and OSC-native DAWs

2.4 DAW SDK — ReaScript (Reaper), Max for Live (Ableton), Control Surface Protocol

3. Supported Plugins — Priority Roadmap

FabFilter — HIGH PRIORITY

Plugin	Key Parameters Mapped
Pro-Q 3	24 bands: freq, gain, Q, type, phase mode
Pro-C 2	Threshold, ratio, attack, release, knee, style

Pro-L 2	Threshold, gain, lookahead, true peak, oversampling
Pro-MB	6 bands: range, threshold, ratio, attack, release
Pro-R	Decay, room size, character, brightness
Saturn 2	6 saturation bands: drive, tone, level, mix

iZotope — HIGH PRIORITY

Plugin	Capabilities
Ozone 11	Maximizer threshold, IRC mode, EQ, imager width, LUFS target
Neutron 4	Per-track EQ, compressor, transient shaper, exciter
RX 10	Noise reduction, de-click, dialogue isolation

Waves — MEDIUM PRIORITY

SSL G-Master Buss · API 2500 · CLA-76 · H-Reverb · Renaissance EQ · L3-16 · Kramer HLS

Universal Audio — MEDIUM PRIORITY

1176 · LA-2A · Neve 1073 · API Vision · Ocean Way Studios

Native Instruments — MEDIUM PRIORITY

Massive X · Kontakt 7 · Guitar Rig 7 · Supercharger GT

4. ParameterMapper — Core Integration Engine

Responsibilities:

- DISCOVERY — Scans all exposed parameters at plugin load
- NAMING — Maps numeric IDs to semantic names (47 → "Band3_Gain")
- SCALING — Converts physical values to normalised (−3dB → 0.42)
- VALIDATION — Checks range and type before applying
- UNDO — Saves prior state for instant rollback

5. Plugin Database Format

```
{
  "plugin_id": "fabfilter_proq3",
  "display_name": "FabFilter Pro-Q 3",
  "vst3_guid": "E8E38A93-21B0-4B8D-B7EC-B54B8B9CF48C",
  "parameters": [
    {
      "id": 0, "name": "Band1_Frequency",
      "alias": ["freq1", "eq band 1 freq"],
      "min": 20.0, "max": 20000.0,
```

```
    "scale": "logarithmic", "unit": "Hz"
  }
],
"ai_presets": {
  "remove_mud": { "Band2_Freq": 220, "Band2_Gain": -3.5, "Band2_Q": 2.1 },
  "add_presence": { "Band4_Freq": 3500, "Band4_Gain": 2.0 }
}
```

6. Safety Constraints

- No action applied without explicit user confirmation (unless auto-apply enabled)
- Full undo stack maintained for every action
- No gain changes > 12dB in a single command
- No modifications during recording (unless override set)
- No audio routing changes (feedback risk)

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